

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 33

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Natural pressure of gas above the oil (or dissolved in it) or water beneath forces oil up the production well; pump-jacks ('nodding donkeys') increase flow • Pressure falls as oil is removed so the remaining oil no longer flows to the well	2
01.2	• Water or natural gas is pumped down an injection well to maintain the pressure and flow of oil (recovery to about 40%) • Captured CO₂ can be pumped down instead, increasing oil recovery while storing the CO₂ underground	2
01.3	<i>Any 3 from:</i> • Steam injection (from combustion or solar parabolic concentrators) to heat the oil and reduce its viscosity • Controlled underground combustion to heat the oil • Detergents / solvents to reduce surface tension so oil flows more easily • Bacteria that partially digest heavy oil into lighter oils and produce CO₂ to maintain pressure	3
01.4	• Many wells can be drilled from a single platform, reducing land take / number of rigs • Deposits under locations where rigs cannot be placed (urban areas, sensitive habitats) can be reached • Drilling can follow softer strata (quicker) and target multiple small reservoirs up to 10 km away, increasing total recovery	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187-188; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
02.1	- Visible light passes through the atmosphere easily and is absorbed by the Earth's surface, which warms and emits infrared - Greenhouse gases (naturally water vapour and CO₂) absorb infrared better than the average for the atmosphere, converting it to heat and re-emitting some back to the surface - This keeps the surface about 33 °C warmer than it would otherwise be	3
02.2	- CO₂ is emitted in far larger quantities (every combustion process, deforestation) so its total effect is greatest - CO₂ has a long residence time so it accumulates; methane has a short residence time (12 years) so it is removed faster - CFC emissions are now very small following the Montreal Protocol	3
02.3	- Methane has a much higher global warming potential (28x) so each tonne removed has a bigger effect - Its short residence time (about 12 years) means atmospheric concentrations fall quickly once emissions are cut	2
02.4	<i>Any 2 from:</i> - Collect landfill / anaerobic digester gas and use as fuel - Changes in cattle diet / fewer cattle / less meat consumption - Draining rice paddies periodically - Anaerobic digestion of manure with gas capture	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-90; AQA spec 3.2.1.2.	
03.1	- Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium - Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet - International agreement on ownership of the seabed and its resources is needed	3
03.2	- Collectors disturb the seabed and kill benthic organisms living on it - Separating nodules from sediment creates plumes that increase turbidity with unknown consequences - Re-deposition of sediment covers and kills filter feeders and other benthic organisms; slow recovery in the deep sea	3
03.3	- Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) - Exploitable land deposits are limited and mainly in China; chemical separation is complex and produces large amounts of toxic waste - Bacterial adsorption: dissolved rare earth ions adsorb onto the cell surfaces of bacteria (<i>Bacillus subtilis</i>) and are washed off at specific pHs to separate them - Deep sea sediments contain relatively high concentrations but no viable extraction method yet	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.	

Question	Marking guidance	Marks
04.1	• Asbestos: fibres become airborne when it disintegrates and are inhaled, causing asbestosis (scar tissue thickens alveoli) and mesothelioma; disposal: double-wrapped in heavy-duty polythene in a specialised, recorded landfill (secure, permanent, sealed storage) • Cyanide: very toxic enzyme inhibitor (aerobic respiration) if inhaled / ingested; disposal: incineration so carbon and nitrogen are separated and oxidised to CO₂ and NO_x (much smaller problem) • High-level radioactive waste: intense ionising radiation, heat generation, long half-lives; disposal: vitrification - mixed with molten glass in sealed stainless steel containers in ventilated concrete stores	6
04.2	• The fibres are bound / cannot become airborne so cannot be inhaled	1
04.3	• Waste is mixed with a cement slurry and poured into containers of impermeable, unreactive material • The solid cement immobilises the waste so it cannot leak / leach out	2
04.4	• Even if the glass shatters the waste stays encapsulated in the fragments; glass is more stable / resistant to heat and leaching over very long timescales	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 284-285, 303; AQA spec 3.4.3.2.12.	
05.1	• Net held with its straight edge on the river bed facing the current (open to the flow) • The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) • Same substrate type, water depth and flow rate at both sites; same net mesh; repeat samples; invertebrates identified with a key and counted	3
05.2	<i>Any 3 from:</i> • Difficult to standardise the vigour / area kicked • Buried organisms or those fixed to stones are not dislodged • Some organisms swim and avoid the net • Flow rate varies and affects capture • Fragile organisms destroyed	3
05.3	• A fixed area of bed is enclosed by a frame so the sampled area is standardised; side nets reduce escapes • But only works in fast enough flow and is hard to set on rough substrates / deep water	2
05.4	• Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event • Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	

Question	Marking guidance	Marks
06.1	• Double hull: a gap of up to 2 m between the hulls so damage to the outer hull in a collision / grounding does not release oil • Inert gas system: cooled engine exhaust gas (no oxygen) fills the space above the oil so an explosive oil vapour / air mixture cannot form • Separate ballast tanks: ballast water is not carried in oil tanks so no residual oil is discharged when ballast is pumped out • Twin engines / rudders: if one fails the ship can still be steered / propelled and is not carried onto rocks by currents or wind • Crude oil washing: sludges are mixed and removed with the cargo so tanks do not need washing with seawater that was then discharged; oily water is unloaded at the terminal	5
06.2	• Offshore routes keep tankers further from the coast so if there is engine failure there is more time to deal with it before the ship is carried ashore • GPS gives accurate navigation; AIS lets ship positions be monitored and coordinated to avoid collisions and groundings	2
06.3	• The tank stands on an impermeable base surrounded by a wall • The volume enclosed by the wall is greater than the tank volume so all the oil is contained if the tank splits / cannot overflow	2
06.4	• Separates floating oil from drainage water so the oil is retained (for later treatment) and only water flows to the drains / river	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 259-260; AQA spec 3.4.3.2.7.	
07.1	• $(120 + 95) - (2 + 45) = 168 \text{ t ha}^{-1}$	1
07.2	• Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO₂ • Harvest removes biomass so less dead organic matter is returned; pasture roots and litter constantly add organic matter	2
07.3	<i>Any 2 from:</i> • Low / zero tillage • Adding manure, compost or mulch • Growing cover crops / leaving crop residues • Converting to long-term grass or tree crops in rotation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	
08.1	• $180 / 1.2 = 150$	1
08.2	• Cadmium and zinc occur together in the same ores so mine drainage / smelting releases both	1
08.3	• Mine drainage water contaminated irrigation water / paddy soil so rice absorbed cadmium ($1.2 \text{ vs } 0.05 \text{ mg kg}^{-1}$) • Cadmium is liposoluble / not excreted so it bioaccumulates in the body (kidneys) with chronic exposure to small doses over years • Cadmium causes kidney failure and bone decalcification / skeletal collapse (itai-itai disease) • Rice is the staple food so residents receive a large dose; village B has no exposure and no disease (correlation supports causation)	4

Question	Marking guidance	Marks
08.4	<i>Any 2 from:</i> • Collect mine drainage water and neutralise / raise pH with lime so cadmium precipitates and stays insoluble • Cover / cap spoil heaps to prevent leaching and dust • Recycle nickel-cadmium batteries / hazardous waste landfill • Phytoremediation of contaminated paddy soils	2
08.5	• Manufacture may release cadmium in effluents / dust • Panels have a limited life; disposal of old panels (landfill, incineration of pigments) could release cadmium unless recycled	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
09.1	• Soil loss increases as vegetation cover decreases: forest lowest (0.5), bare soil highest ($60 \text{ t ha}^{-1} \text{ yr}^{-1}$) - over 100 times more • Ploughing downslope loses nearly three times as much as contour ploughing for the same crop	2
09.2	<i>Any 4 from:</i> • Leaves / leaf litter intercept raindrops so soil particles are not dislodged by rain splash • Roots bind soil particles together • Vegetation acts as a windbreak reducing wind velocity and the kinetic energy available to carry particles • Plants increase infiltration (root channels, organic matter) so less surface runoff • Organic matter / humus from vegetation binds particles into stable peds	4
09.3	• Furrows running downslope act as channels / gulleys that increase the velocity (kinetic energy) of runoff • Furrows along the contour trap the water so it loses energy, deposits particles and infiltrates	2
09.4	• In the dry season the soil (especially low-clay soils) is loose and dry with little cohesion so wind blows it away • In the wet season heavy rain causes rain splash and surface runoff when infiltration capacity is exceeded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 162-164; AQA spec 3.2.5.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Principles: dB is logarithmic ($10 \text{ dB} = 10x$; $3 \text{ dB} = 2x$) so small reductions in dB are large reductions in intensity; effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • Aircraft design: high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • Aircraft operation: take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • Airport design / location: away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • Railway: track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • Road: porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • Industrial: silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • Domestic: acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • Evaluation: which approaches are most effective / cost-effective; conflicts (airports near cities for convenience); legislation and monitoring (Leq 57 dB contours) needed to enforce	9
10.2	• The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Stages of investigation: identifying the issue (Halley Station UV data, Dobson unit measurements), hypothesis formation (Rowland-Molina 1974: persistence, UV dissociation, chlorine reactions), data collection (ground, satellite, balloon and aircraft air sampling detecting ClO), analysis of spatial, temporal and altitude variability, proposal of solutions, enactment • Why Antarctica: low stratospheric temperatures, ice crystals, stratospheric clouds, polar vortex, spring sunlight • Effects of UVB: skin cancer, cataracts, DNA damage, plant and marine damage - motivating action • Solutions: Vienna Convention, Montreal Protocol - phase-out and bans, alternative processes (pump sprays), alternative materials (HCFCs, HFCs, hydrocarbons, HFAs), collection and incineration of CFCs, fund for developing countries • Evaluation of success: ozone hole area / depth stabilising, CFC levels falling; limitations - persistence, remaining ODSs, HFCs as greenhouse gases, decades to full recovery • Comparison with other atmospheric problems (climate change, acid rain) - why ozone was easier; role of peer review, international cooperation, availability of substitutes • Conclusion on the extent to which it is a model for other issues Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-113; AQA spec 3.2.1.3.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use • Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	25